



Naga Sai Krishna Kesani

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Profile Summary

- Over 5 years of experience building and deploying end-to-end machine learning and deep learning solutions to solve complex business problems across industries.
- Proficient in **Python, R, and SQL** for data preprocessing, feature engineering, and model development using frameworks like **TensorFlow, PyTorch, and scikit-learn**.
- Experienced in designing predictive systems across **NLP, computer vision, and time-series forecasting** using algorithms such as **XGBoost, LSTM, and ARIMA**.
- Skilled in deploying ML models on **AWS SageMaker, Google Cloud, and Azure ML**, and automating workflows with **Docker, Kubernetes, Airflow, and MLflow**.
- Adept at visualizing and communicating insights using **Tableau, Plotly, and Matplotlib**, and applying **Agile and Git-based** development best practices.

Education

Rochester Institute of Technology

Master of Science in Data Science (GPA: 3.8/4.0)

Rochester, NY

May 2025

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Mechanical Engineering (GPA: 4.0/4.0)

Bengaluru, India

May 2022

Experience

MFS Investment Management

Machine Learning Engineer

Boston, Massachusetts

July 2025 – Present

MFS Investment Management is a leading global asset management firm providing investment solutions to financial advisors, institutions, and intermediaries worldwide. As an ML Engineer, the role focuses on developing scalable, data-driven models to enhance investment insights and distribution efficiency.

Environment: Python, SQL, Snowflake, Azure ML, Scikit-learn, XGBoost, Tableau, MLflow, Airflow, Git.

- Developed the “**Note Quality Metric**”, an ML-driven framework to classify wholesaler meeting notes as good or bad, converting probabilistic outputs into an intuitive **1–10 rating scale**.
- Built an **NLP pipeline** with **NLTK, SpaCy, TextStat, and VADER** for preprocessing, sentiment analysis, and feature extraction, enabling structured and actionable note evaluation.
- Trained and validated an **XGBoost classifier** on 800 labeled/semi-labeled notes with **5-fold cross-validation**, optimizing hyperparameters for balanced accuracy, precision, and recall.
- Generated **actionable feedback suggestions** (e.g., follow-ups, product mentions, clarity improvements), improving documentation consistency and reducing subjectivity in client interactions.
- Collaborated with distribution strategists and leveraged **Snowflake, PyCharm Pro, and Bitbucket Git** to deliver scalable, data-driven insights supporting sales and investment teams.

State Street Corporation

Machine Learning Engineer

Boston, Massachusetts

Nov 2023 – Jul 2025

State Street Corporation is a leading global financial services and bank holding company offering investment management, research, and analytics solutions to institutional investors. As an ML Engineer, the role involves designing and deploying machine learning models for advisor propensity modeling and fund distribution analytics.

Environment: Python, Snowflake, Azure ML, Scikit-learn, XGBoost, SHAP, Tableau, Airflow, MLflow, SQL.

- Automated end-to-end **data ingestion and preprocessing pipelines** using **Python (Pandas, NumPy, SQLAlchemy)** and **Apache Airflow**, ensuring real-time data availability for model training and accelerating product insights by 30%.
- Leveraged **Snowflake** and **Azure Databricks** to engineer advisor-level features from multi-source datasets (**transactions, AUM, entity, and interactions**), creating a unified data model for downstream ML workflows.
- Applied **Random Oversampling (Imbalanced-Learn)** and **K-fold cross-validation** for class balancing and hyperparameter optimization using **scikit-learn**, improving model robustness and generalization.
- Trained and deployed **XGBoost** models on **Azure Machine Learning** integrated with **Docker** containers, predicting advisor propensity to invest in Target Date Funds with production-ready scalability.
- Conducted **feature importance (SHAP)** and **statistical analysis (ANOVA)** to identify the top 20 predictors of investment behavior, visualized through interactive **Tableau** dashboards and **Plotly** reports.
- Implemented **MLOps automation** using **Airflow, MLflow, and Azure DevOps CI/CD pipelines** for reproducible training, model tracking, and scheduled retraining in production environments.

MathCo
ML Engineer-Data Scientist

Bengaluru, India
Jan 2022 – Jul 2023

TheMathCompany (MathCo) is a global data and analytics consulting firm that builds AI-powered data products and enterprise solutions for Fortune 500 clients across diverse industries. As a Data Scientist, the role focuses on delivering end-to-end data engineering, visualization, and analytics solutions to drive business impact.

***Environment:** Python, SQL, Snowflake, dbt, Tableau, Power BI, PySpark, Airflow, Git, Azure.*

- Engineered automated **data pipelines** using **Google BigQuery**, **Apache Airflow**, and **Pandas**, integrating data from **15+ media platforms** for real-time ingestion, transformation, and centralization, improving reporting **efficiency by 30%**.
- Built and containerized fully automated **ETL workflows** with **Docker** and orchestrated scheduling in **Airflow**, ingesting data from **CDN**, **DAM**, and project management systems into **Snowflake** and **Delta Lake**, ensuring scalable, fault-tolerant pipelines.
- Implemented end-to-end **MLOps pipelines** for data validation, retraining, and deployment using **CI/CD (GitHub Actions, Jenkins)**, **Kubernetes**, and **Azure ML**, achieving a **100% increase in operational efficiency**.
- Developed, containerized, and deployed **machine learning models** for claim frequency and severity prediction via **FastAPI** microservices on **AWS SageMaker**, enabling automated **premium pricing** and improving underwriting risk control.
- Performed large-scale **EDA**, **feature engineering**, and **preprocessing** using **Python (NumPy, scikit-learn, SciPy)** and stored processed data in **Snowflake** and **PostgreSQL** for model training and evaluation.
- Configured continuous model monitoring and versioning with **MLflow**, integrated metrics visualization in **Prometheus** and **Grafana**, and automated retraining triggers using **Airflow**, ensuring robust performance and drift detection.

Alkem Laboratories
ML Engineer-Data Scientist

Mumbai, India
Jan 2021 – Jan 2022

Alkem Laboratories Limited is a leading Indian multinational pharmaceutical company engaged in the development, manufacturing, and marketing of healthcare and biotechnology products. As a Data Scientist, the role focuses on developing deep learning models to analyze healthcare datasets for predictive accuracy and operational insights.

***Environment:** Python, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, SQL, Git.*

- Designed, developed, and deployed **predictive analytics and deep learning models** using TensorFlow, Keras, PyTorch, XGBoost, and LightGBM to enhance healthcare applications and insurance risk portfolios.
- Built and optimized **real-time and batch data pipelines** using Apache Kafka, Flink, Airflow, Hadoop, Spark, and SQL/NoSQL databases (MySQL, PostgreSQL, MongoDB, Cassandra) to process large-scale claims and transactional data, enabling actionable insights and live fraud monitoring.
- Developed **AI-powered automation solutions**, including chatbots and virtual assistants using Dialogflow and Rasa, deploying models through Docker, Kubernetes, Jenkins, and GitLab CI/CD to ensure efficient, production-grade integration.
- Implemented **unsupervised and ensemble learning techniques** (K-Means, DBSCAN, Bayesian Networks) and leveraged AutoML frameworks (**Google Cloud AutoML**, **H2O.ai**) for faster and more accurate model selection, tuning, and deployment.
- Designed and visualized **business intelligence dashboards** using Tableau, Power BI, and Plotly to deliver regulatory-compliant insights and improve decision-making in underwriting, pricing strategies, and claims management.

Technical Skills

Programming Languages: Python, R, Java, Scala, C++, SQL, Julia, Bash/Shell Scripting

Machine Learning Frameworks & Libraries: TensorFlow, PyTorch, Keras, scikit-learn, XGBoost, LightGBM, H2O.ai, OpenCV, CatBoost, FastAI

Deep Learning Models: CNNs, RNNs, LSTMs, GRUs, Transformers, GANs, Autoencoders, Attention Mechanisms

Big Data & Distributed Systems: Hadoop, Apache Spark, Flink, Kafka, Hive, Delta Lake, Dask

Cloud Platforms & Services: AWS (SageMaker, Lambda, Redshift, EC2, S3, Glue); Google Cloud (Vertex AI, BigQuery, Dataflow, Cloud Functions); Azure (Machine Learning Studio, Databricks, Azure Functions, Synapse Analytics)

Data Processing & Analysis: Pandas, NumPy, Dask, NLTK, SpaCy, SciPy, Featuretools

Data Visualization & BI Tools: Matplotlib, Seaborn, Plotly, Dash, Tableau, Power BI

Databases: SQL (MySQL, PostgreSQL, SQL Server), NoSQL (MongoDB, Cassandra, DynamoDB, Redis)

DevOps & MLOps: Docker, Kubernetes, Jenkins, Git, MLflow, Airflow, Argo, Prefect, CI/CD Pipelines

Model Deployment & APIs: Flask, FastAPI, Streamlit, REST APIs, gRPC, TorchServe, TensorFlow Serving

AutoML & Optimization Tools: Google AutoML, H2O AutoML, Azure AutoML, Grid Search, Random Search, Bayesian Optimization, Hyperparameter Tuning

Explainable AI & Responsible AI: SHAP, LIME, Fairlearn, What-If Tool, Captum

Edge AI & Embedded Systems: TensorFlow Lite, ONNX Runtime, Nvidia Jetson, Coral Edge TPU